



Biomedical Signal Processing and Signal Processors

Dr. Mehdi Hasan Chowdhury, Professor, HKPFS Scholar
Chittagong University of Engineering & Technology (CUET)



Abstract:

Biomedical signal processing plays a central role in modern eHealth systems by enabling the acquisition, conditioning, compression, interpretation, and transmission of electrophysiological signals, also known as biosignals. This speech introduces the concept of signal processors and highlights how reconfigurable computing, particularly FPGA-based architectures, can support real-time, multichannel, and domain-specific biomedical applications. Beginning with the fundamentals of signal processing hardware and the motivation for using FPGAs, the talk discusses the state of the art in reconfigurable computing for eHealth, including medical diagnosis, biosignal conditioning, assistive technologies, prosthesis design, and hardware acceleration. The talk then examines three representative biomedical signal processor designs. The first focuses on time-critical ECG compression for long-term recording and remote transmission. The second presents an EMG-based diagnostic system to distinguish neuromuscular conditions, such as myopathy and neuropathy, using computationally efficient signal features. The third explores EEG-based depth-of-anesthesia measurement for monitoring consciousness during experimental and surgical procedures. After that, a few more state-of-the-art works will be explored. Through these examples, the speech demonstrates how biomedical signal processors can translate physiological data into clinically meaningful information while meeting practical constraints of speed, power, reliability, and hardware resources. The talk concludes by identifying future research opportunities in unified biosignal processing platforms, clinically accepted diagnostic devices, and advanced biomedical systems such as neural prostheses and artificial organs.

Biography:

Dr. Mehdi Hasan Chowdhury is a Professor in the Department of Electrical and Electronic Engineering at Chittagong University of Engineering & Technology (CUET), Bangladesh. He received the B.Sc. and M.Sc. degrees in Electrical and Electronic Engineering from CUET, and the Ph.D. degree from the Department of Electrical Engineering at City University of Hong Kong in 2021. He was awarded the prestigious Hong Kong Ph.D. Fellowship Scheme (HKPFS) in 2018. Prof. Chowdhury's research interests include biomedical engineering, digital signal processing, and embedded systems design. He has published more than 50 research papers in international journals and conferences and holds two United States patents. He is a Senior Member of the IEEE, the counselor of the IEEE CUET Student Branch, and an executive member of the IEEE R10 SAC.

3 June 2026 (Wed); 2pm – 4pm; Yeung P1402 + <https://cityu.zoom.us/j/96742093029>